Kai Sheng Tai

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About

I am an AI researcher specializing in model pre-training, LLM inference efficiency, and neural network architecture design.

At Meta, I worked as a technical lead on text embedding model development and on developing efficient LLMs for on-device deployment.

My academic research leverages structure in data to build more efficient and effective models: for instance, by exploiting linguistic structure with Tree-LSTMs, group symmetries with Equivariant Transformers, and the Earth's physical constraints for ML-driven earthquake detection.

Technical Skills

Core Competencies: Deep Learning, LLM Pre-/Post-training, Distributed Training, Parameter Quantization, Neural Network Architecture Design, Data Sketching Algorithms

Frameworks & Tools: Python, PyTorch, Large-Scale GPU Clusters

Professional Experience

Research Scientist, Meta (2021 – 2025)

Meta Reality Labs (2025)

- Optimized inference latency and memory utilization for on-device LLMs (∼1B parameters) by developing advanced parameter quantization and weight pruning techniques.
- Adapted pretrained dense checkpoints to use interleaved local-global attention, significantly reducing KV cache memory usage.
- Engineered mid-/post-training SFT pipelines to expand model capabilities, successfully integrating tool calling and audio processing functionalities.

Modern Recommendation Systems (2022 – 2024)

- Led end-to-end development of a next-generation multilingual text embedding model to serve downstream ranking, recommendation, and safety systems.
- Designed and implemented a text encoder pretraining pipeline, maximizing training efficiency under strict GPU budgets using Fully and Hybrid Sharded Data Parallelism (FSDP/HSDP).
- Partnered with AI Infra & PyTorch teams to optimize deployment throughput, pioneering use of the Nested Jagged Tensor layout to eliminate padding overhead for variable-length sequences.

Facebook AI Applied Research (2021 – 2022)

- Designed a differentiable parameterization of sparse layers that enables end-to-end training of sparse CNN and Transformer models, outperforming prior pruning techniques in accuracy.

Advising & Mentorship (throughout)

- Co-advised multiple Ph.D. research intern projects on reinforcement learning with verifiable rewards (RLVR), multi-entity video transformers, and unsupervised prompt optimization.

Senior Data Scientist, MetaMind (acquired by Salesforce) (2015 – 2016)

- Led architecture development and training of 3D convolutional neural networks specialized for medical MRI classification prior to the company's acquisition.

Education & Academic Research

Ph.D. in Computer Science, Stanford University (2016 – 2021)
Thesis: *Statistical Machine Learning Under Resource Constraints*
Advisors: Peter Bailis and Gregory Valiant

- Designed and published efficient algorithms for machine learning under stringent resource constraints, focusing on sketching, compression, and semi-supervised classification.
- Developed Equivariant Transformer Networks to incorporate continuous transformation groups into neural architectures, utilizing structural symmetries to improve model robustness.
- Created novel data sketching algorithms for streaming environments, including linear classifier sketching and moment-based quantile sketches for high-cardinality aggregation queries.

M.S. in Computer Science, Stanford University (2013 – 2015)

- Developed the Tree-Structured Long Short-Term Memory (Tree-LSTM) architecture, demonstrating that explicit tree-structured topologies for hierarchical information processing can outperform sequential RNNs on semantic representation tasks.

A.B. in Physics, *magna cum laude* (high honors), Princeton University (2009 – 2013)
Senior Thesis: Detecting Gravitational Waves from Highly Eccentric Compact Binaries. Advisors: Frans Pretorius, Sean McWilliams.

GPA: 3.92/4.00, Phi Beta Kappa

Selected Publications

Zhu, H., Zhang, Z., Huang, H., Su, D., Liu, Z., Zhao, J., Fedorov, I., Pirsiavash, H., Sha, Z., Lee, J., Pan, D., Wang, Z., Tian, Y., and **Tai, K.S.**. The Path Not Taken: RLVR Provably Learns Off the Principals. *arXiv 2025*.

Huber, P., Chang, E., Wen, W., Fedorov, I., Elgamal, T., Huang, H., Suda, N., Sankar, C., Vogeti, V., Wang, Y., Gladkov, A., **Tai, K.S.**, Elogeel, A., Hefny, T., Chandra, V., Aly, A., Kumar, A., Krishnamoorthi, R., and Sagar, A. MobileLLM-Pro Technical Report. *arXiv 2025*.

Nguyen, A.T., **Tai, K.S.**, Chen, S., Shukla, S., Yu, H., Torr, P., Tian, T., and Lim, S. uCAP: An Unsupervised Prompting Method for Vision-Language Models. *ECCV 2024 (oral presentation)*.

Tai, K.S., Tian, T., and Lim, S. Spartan: Differentiable Sparsity via Regularized Transportation. *NeurIPS 2022*.

Zhu, W.*, **Tai, K.S.***, Mousavi, S.M., Bailis, P., and Beroza, G. An End-to-End Earthquake Monitoring Method for Joint Earthquake Detection and Association using Deep Learning. *Journal of Geophysical Research: Solid Earth*, 2022. (*equal contribution)

Tai, K.S., Bailis, P., and Valiant, G. Sinkhorn Label Allocation: Semi-Supervised Classification via Annealed Self-Training. *ICML 2021*.

Tai, K.S., Bailis, P., and Valiant, G. Equivariant Transformer Networks. *ICML 2019*.

Tai, K.S., Sharan, V., Bailis, P., and Valiant, G. Sketching Linear Classifiers over Data Streams. *SIGMOD 2018*.

Tai, K.S., Socher, R., and Manning, C.D. Improved Semantic Representations from Tree-Structured Long Short-Term Memory Networks. *ACL 2015*.